ARCHITECTURAL PROPOSAL | ZERO CODE MODIFICATION PHASE | MERN + CLOUD LLM

MITRA Employability Portal

Complete AI Services & LLM Proposal — Architecture, Feasibility, Cost Governance & Roadmap

AUTHOR / TARGET	ARCHITECTURE	HOSTING TARGET	STATUS
MITRA AI Working Group	React 18 + Express 5 + MongoDB	Vercel (Client) + Render (API)	Approved Blueprint / Pre-Implementation

Core Implementation Principle

Traditional Processing First, LLM Last. The MITRA system adheres to strict psychological validity and high cost efficiency. All scoring, proctoring, psychometric math, and clean PDF extraction run deterministically within Node.js. Generative LLMs are invoked selectively for question generation, unstructured text transformation, and personalized feedback synthesis.

1. AI Question Generation Strategy

Each cognitive assessment domain in MITRA imposes different demands on generative models:

ASSESSMENT DOMAIN	COGNITIVE / STRUCTURAL CHALLENGE	PRIMARY RECOMMENDED ENGINE	OPERATIONAL STRATEGY
Quantitative Aptitude	Calculation drift & symbolic arithmetic errors in LLMs	**Gemini 2.5 Flash / Groq** (T: 0.1)	Force step-by-step mathematical reasoning in internal rationale before outputting final options.
Logical & Analytical	Constraint satisfaction, syllogisms, and puzzle consistency	**Gemini 2.5 Flash**	Calibrated template validation; option shuffling with guaranteed single truth value.
Verbal & Communication	Grammatical precision, nuance, reading passages	**Groq (Llama-3.3-70B)**	Ultra-fast token generation for reading comprehension and linguistic nuance.
Domain Technical (CS/Engg)	Syntax accuracy, modern frameworks, executable code	**Gemini 2.5 Flash**	Enforce strict code snippet blocks and exact 4-choice responseSchema.
Situational Judgment (SJT)	Plausible workplace behavioral dilemmas	**Gemini 2.5 Flash / Groq**	Formulate professional scenarios with 4 graded behavioral alternatives.
Psychometric Scale Items	Psychometric test-retest reliability & construct validity	**Deterministic Item Bank (No LLM)**	Do NOT generate items dynamically during live tests; use pre-standardized Likert scales.
PDF-Derived Questions	Document grounding and noise extraction	**Hybrid: pdf-parse + Gemini**	Extract text locally; LLM parses only ambiguous or unstructured question blocks.

2. PDF Question Extraction: Traditional vs. LLM Comparison

Directly feeding multi-page PDFs to cloud multimodal APIs consumes tens of thousands of tokens per upload. MITRA utilizes a **6-stage hybrid extraction pipeline** that minimizes token expenditure:

METRIC	TRADITIONAL EXTRACTION (PDF-PARSE)	PURE LLM (DIRECT PDF MULTIMODAL)	RECOMMENDED MITRA HYBRID PIPELINE
API Token Cost	$0.00 (Runs locally in Node.js)	$0.05 - $0.25 per uploaded test	**Near $0.00** (LLM only for ambiguous blocks)
Execution Latency	Sub-second (100–300 ms)	8–25 seconds per document	**1–3 seconds** end-to-end
Standard MCQ Parsing	100% accurate via regex	Prone to minor formatting variations	**Deterministic & auditable**
Complex / Scrambled Text	May fail on broken multi-column layouts	High semantic reconstruction ability	**LLM activated only on low regex confidence**

The Recommended 6-Stage Hybrid Flow

PDF Upload → 1. Buffer Extraction (pdf-parse) → 2. Text Normalization → 3. Deterministic Regex Extractor → *(If clean ≥ 90%, bypass LLM; else **4. Chunked Micro-LLM Structuring**)* → **5. Schema Validation → 6. MongoDB Question Bank**.

3. Psychometric Assessment: Deterministic Logic vs. AI Value-Add

Psychometric evaluations must deliver objective, legally defensible, and reliable career guidance. Scoring is kept strictly mathematical; LLMs are utilized solely for synthesis.

100% Deterministic (Node.js)

• Likert scale scoring (1 to 5)
• Reverse-score transformations
• Big-Five & RIASEC aggregations
• Employability Readiness Index (ERI)
• Normative cohort percentile lookup

Selective Generative LLM

• Executive narrative synthesis
• Translating numerical trait profiles into 3 actionable workplace strengths
• Identifying personal blind spots
• Targeted career pathway recommendations

Data Sent to LLM

• No raw student questionnaire records
• Only 5 calculated trait percentages sent as a compact JSON object (~150 tokens)
• 85% token savings vs full transcript

4. Resume Analysis Engine

MITRA performs dual-layer resume evaluation combining deterministic parsing with semantic LLM audit:

- **Layer 1 (Local Extraction):** Text is extracted from PDF/DOCX via `pdf-parse`. Section headers (Education, Experience, Projects, Skills) are identified via regex patterns.
- **Layer 2 (Semantic Analysis via Gemini 2.5 Flash):** The structured sections (~1,200 tokens) are evaluated for:
  - **Google X-Y-Z Impact Metric:** Checks whether bullet points demonstrate "Accomplished [X], as measured by [Y], by doing [Z]".
  - **Departmental Gap Analysis:** Flags missing foundational technologies for the candidate's target branch.
  - **Actionable Feedback:** Returns structured JSON with ATS score (0-100), top 3 strengths, critical omissions, and recommended keywords.
- **Cost Profile:** Less than $0.0005 per analyzed resume. 1,000 student submissions cost less than $0.50 total.

5. AI Interview Preparation & Communication Analysis

The interview coaching engine uses a two-tier model to balance responsiveness with comprehensive grading:

- **Dynamic Turn-by-Turn Interviewer:** Powered by **Groq Cloud (Llama-3.3-70B)**. Answers arrive within 400ms, maintaining natural interview cadence. The model analyzes student responses and probes deeper into architectural trade-offs or behavioral actions.
- **Comprehensive Multi-Rubric Evaluation:** Upon interview completion, the complete transcript is analyzed by **Gemini 2.5 Flash** across 7 standardized rubrics: *1. Grammar & Syntax, 2. Fluency &*

Coherence, 3. Vocabulary Precision, 4. Relevance, 5. STAR Structure, 6. Clarity, 7. Professional Confidence Indicator.

6. AI Chatbot / Employability Assistant: Feasibility & RAG

- **Verdict: Optional for MVP; Valuable Post-MVP as a Guided Assistant.**
- **Direct LLM vs. RAG:** Direct prompting of an ungrounded LLM leads to hallucinations regarding college-specific placement cutoffs, exam dates, and attendance policies. A true MITRA assistant must use **Retrieval-Augmented Generation (RAG)**.
- **Knowledge Grounding:** Institutional placement policy handbooks, department course syllabi, assessment schedules, and past company hiring criteria.

7. Comprehensive AI Provider & Model Benchmark

PROVIDER & MODEL	MCQ / REASONING	STRUCTURED JSON	INFERENCE SPEED	FREE TIER / QUOTA	RENDER / VERCEL FIT	VERDICT FOR MITRA
Google Gemini (2.5 Flash)	Very High (MMLU 88%+)	Native Schema Mode	Fast (~120 tps)	Generous AI Studio tier	100% Native (Node SDK)	**PRIMARY WINNER**
Groq (Llama-3.3-70B)	High (Open Source SOTA)	Strict JSON Object	Ultra-Fast (~300 tps)	Generous developer tier	100% Native (Axios/REST)	**BACKUP & REAL-TIME**
OpenAI (GPT-4o-mini)	Very High	Native Schema Mode	Moderate (~80 tps)	Prepaid balance only	100% Native (Node SDK)	Commercial Alternative
Anthropic (Claude 3.5 Haiku)	Exceptional	Tool Calling / Prompted	Moderate (~70 tps)	No free tier	100% Native (Node SDK)	High Cost / Overkill
Hugging Face Inference	Variable by model	Unreliable formatting	Slow / Cold starts	Frequent 503 / timeouts	Partial (Frequent aborts)	Emergency Fallback Only
Self-Hosted (Ollama/vLLM)	Depends on VRAM	Requires guidance	Hardware limited	Hardware costs ($50+/mo)	Impossible on Render Free	**NOT Recommended**

Is a Separate Python AI Backend Required?

NO. Both Gemini (`@google/genai`) and Groq (REST / Axios) feature native Node.js support. PDF text extraction (`pdf-parse`) executes in Node.js. A separate Python backend would add unnecessary deployment complexity, double memory overhead on Render, and introduce redundant inter-service authentication.

8. Recommended MITRA System Architecture

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|                                     MITRA ARCHITECTURAL TIERS                                    |
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| TIER A: DETERMINISTIC CORE (MERN Stack - Zero AI Latency & Cost)                                 |
|   - Authentication & Role Middleware (Student / Admin / Recruiter)                               |
|   - Assessment Delivery, Proctoring Tracker (Tab switches, fullscreen events)                    |
|   - Psychometric Likert Scoring, ERI Aggregation, and Cohort Benchmarks                          |
|   - Local PDF ASCII/Unicode Text Extraction via pdf-parse                                        |
|   - Deterministic Regex MCQ Pattern Extractor                                                    |
|   - Strict Schema Validator & DB Upsert Engine                                                    |
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| TIER B: TARGETED AI CLOUD ENGINE (Structured Services via Gemini 2.5 Flash / Groq)               |
|   - Admin Dynamic Question Bank Generator (Batched in 10-item chunks)                            |
|   - Unstructured PDF Question Structuring (Triggered only on low regex confidence)               |
|   - Psychometric Executive Narrative & Growth Roadmap Synthesizer                                |
|   - Resume ATS Semantic Analyzer & Missing Keyword Recommender                                   |
|   - Dynamic Interview Scenario & Adaptive Follow-up Engine                                        |
|   - Multi-Rubric Communication Evaluator (Grammar, Fluency, STAR, Relevance)                     |
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| TIER C: POST-MVP AI EXPANSION (Deferred)                                                         |
|   - Institutional Policy & Syllabus RAG Assistant (Vector Search + LLM)                          |
|   - Speech-to-Speech Real-Time Vocal Interviewer                                                 |
|   - Automated Code Execution & AST Static Analysis Grader                                        |
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9. Cost & Token Optimization Strategy

- **Local Pre-Filtering:** If a PDF matches standard MCQ regex patterns with ≥90% accuracy, zero tokens are dispatched to the LLM.
- **Pre-Computed Trait Vectors:** For psychometrics, raw question responses are never sent to the LLM. Only 5 trait percentages (~150 tokens) are submitted.
- **Deduplication Caching:** Generated questions are fingerprinted via SHA-256 hashes and cached in MongoDB to eliminate duplicate generation expenses.
- **Chunked Batch Processing:** Generation occurs in small batches (max 10 items) with timeouts to avoid output token cutoffs.
- **Three-Stage Fallback Ladder:** Primary AI (Gemini) → Secondary AI (Groq) → Grounded Academic Fallback Bank (100% uptime guarantee).

Executive Summary & Final Architectural Verdict

Primary AI Provider	**Google Gemini (Gemini 2.5 Flash)** — Unmatched structured JSON, generous quota, Node.js SDK
Backup AI Provider	**Groq Cloud (Llama-3.3-70B)** — Ultra-low latency (<400ms), OpenAI-compatible REST API
PDF Extraction Engine	**Local `pdf-parse` + Regex** (Hybrid LLM fallback only for complex layouts)
RAG Implementation	**Optional (Post-MVP)** — Not required for core assessments or psychometrics
Separate Python Backend	**Not Required** — Unified MERN Express architecture on Render is optimal
MVP AI Features	1. Admin AI Question Generator (Aptitude & Domain) 2. Hybrid PDF Question Extractor & Structurer 3. Psychometric Narrative & Action Roadmap Synthesizer 4. Dynamic Interview Simulation & Communication Evaluator
Future AI Features	1. Campus RAG Policy Assistant • 2. Speech-to-Speech Vocal Interview • 3. AST Code Grader